

Flexible and universal autofocus based on amplitude difference of fractional Fourier transform

Ziyang Li^a, Xuyang Zhou^a, Yiran Wang^a, Guancheng Huang^a, Shutian Liu^a, Bin Gao^b, Zhengjun Liu^{a,*}

^a School of Physics, Harbin Institute of Technology, Harbin 150001, China

^b College of Data Science and Technology, Heilongjiang University, Harbin 150080, China

ARTICLE INFO

Keywords:

Coherent diffraction imaging
Autofocus algorithm
Clarity evaluation function
Phase retrieval

ABSTRACT

Autofocus technology is a crucial technique in computational optical imaging. The clarity evaluation function (CEF) serves as the criterion for autofocus, making it the core of the autofocus algorithm. In this paper, we propose a CEF based on amplitude differences of fractional Fourier transform (ADFrFT). ADFrFT extracts information from the fractional domain, which corresponds to the fusion of the spatial and frequency domains. This renders ADFrFT highly accurate and robust to noise, making it suitable for various types of sample images. Additionally, ADFrFT provides an added dimension of flexibility to address different autofocus requirements for distinct sample image types and focusing scenarios. Simulations and experiments confirmed the effectiveness of ADFrFT, and we improved the imaging resolution via distance calibration in the coaxial multi-distance coherent diffraction imaging (MDCDI) experiment.

1. Introduction

Coherent diffraction imaging is a non-interferometric microscopy technique and an essential method for observing microscopic details and structures of sample objects [1–9]. One of the most critical challenges in this method is determining the distance between the sample plane and the detector plane of the imaging system [10–15]. Distance measurement methods can be categorized into two main approaches: external hardware-based measurements and image information analysis. Physical ranging measurement using external hardware is often limited by the size of the optical system and the precision of the measurement system. Alternatively, autofocus technology based on image information uses standard autofocus algorithms, computer algorithms, and image processing techniques to analyze and process image features. This approach dynamically adjusts the imaging system's focal length and object distance or automatically adjusts the position and focal length of the lens. The primary goal of autofocus technology is to keep the sample image clear and sharp in real-time [16–20].

Autofocus algorithms typically use the clarity evaluation function (CEF) as the focusing criterion [21–23]. CEF assesses the sharpness of an image based on its spatial or frequency domain information. The amplitude is a fundamental characteristic in the spatial domain, and the

integral amplitude (AMP) can be used as the CEF [24]. Clear images generally exhibit a larger grayscale difference compared to blurred images, resulting in a higher variance, so the variance (VAR) can serve as a CEF [25]. CEFs based on the spatial domain are robust against noise, but they fail to provide accurate results when applied to phase-type images with low and evenly distributed amplitude. In addition to the spatial domain, the frequency domain of the image is also critical. The evaluation criteria focus on using first and second-order differentials of the image and employ differential operators such as Roberts, Sobel, and Laplacian operators to extract edge gradient information. The images with sharper edges are considered clearer. Several CEFs have been proposed, including the summed gradient (GRA), summed Laplacian (LAP), squared gradient (SG), Brenner gradient, and Tenengrad gradient [26,27]. Besides edge information, the overall gradient distribution can also serve as an evaluation criterion. By utilizing a two-dimensional Fourier transform, the images with more high-frequency content are considered clearer [28]. However, the frequency-based CEFs tend to be volatile. The presence of multiple local extrema in the autofocus curve can make it difficult to conduct a comprehensive global search. Several precise and feasible CEFs have also been suggested, such as the Tamura coefficients (TC), the Gini index (GI), Tamura and Gini of the gradient (ToG and GoG), summed weighted power spectrum (SPEC) and pixel

* Corresponding author.

E-mail address: zjliu@hit.edu.cn (Z. Liu).

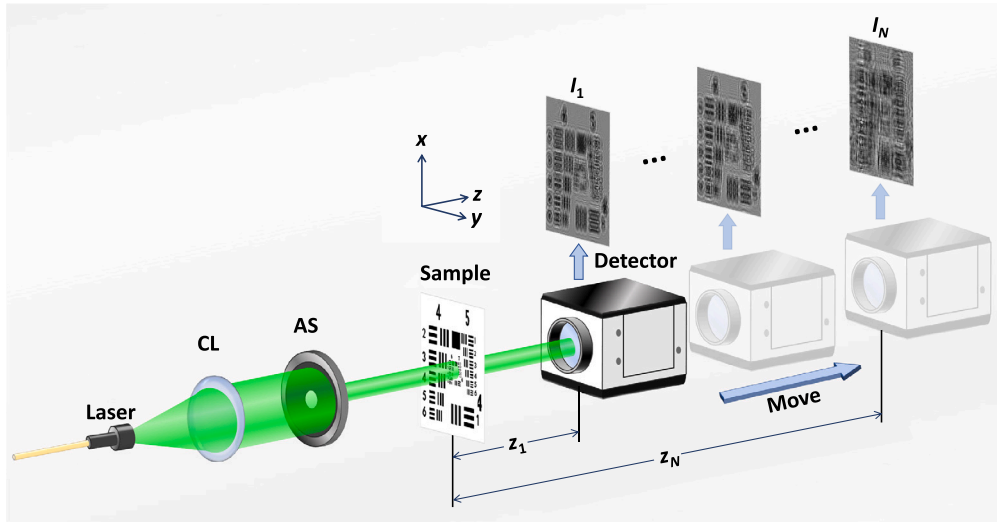


Fig. 1. Experimental setup for coaxial MDCDI system. CL: condenser lens; AS: aperture stop.

difference with the Tanimoto coefficients (PDTC) [26,29–32]. However, it is essential to note that CEFs are designed for particular situations and not be universally applicable.

The conventional CEFs are not the ideal solution for handling various types of sample images since the selection of spatial or frequency domain is fixed for all the CEFs, and their attributes cannot be altered. In this article, we propose a flexible and universal autofocus algorithm to overcome this challenge based on the fractional Fourier transform (FrFT) amplitude difference. Our algorithm operates in the fractional domain, which combines both spatial and frequency domains, and features an adjustable parameter to switch between the domains as needed. This adaptability and versatility enable the algorithm to be used in different sample images and focusing scenarios. After obtaining the diffraction intensity information, we reconstruct the complex hologram of the sample image by using a multi-parameter phase retrieval algorithm [33].

2. Method

2.1. Image clarity quantification

Fig. 1 depicts the coaxial multi-distance coherent diffraction imaging (MDCDI) system [3]. MDCDI utilizes the diffracted intensity distribution of the light field at multiple distances, which is computationally solved to image the original sample. A fiber-coupled laser is condensed and directed to illuminate the sample. The detector collects intensity $I_1, I_2, \dots$, and I_N of the diffraction images at N different distances, which is determined by the complex amplitude field $E_n(x, y; z_n)$ on the n th measurement plane ($n = 1, 2, \dots, N$). In the MDCDI system, the process of solving for the spatial distribution of the light field is achieved by phase retrieval. The multi-distance intensity information is used as an input to the multi-parameter phase retrieval algorithm, and the complex amplitude of the light field in the sample plane is reconstructed using dynamic linear combination phase retrieval (DLCPR) algorithm [33].

Determining the diffraction distance can be done by traversing the optical wave field of the sample image at different propagation distances and calculating the CEF at each distance. The optimal CEF value represents the best plane, and the corresponding propagation distance can be obtained. Noise robustness, axial resolution, and other factors are crucial when assessing CEFs [34,35]. Spatial-based CEFs are ideal for samples with high-frequency noise, while frequency-based CEFs are more suitable for samples with low and uniform amplitude distribution. In environments with a vast search field, frequency-based CEFs

can produce numerous local maxima, making a global search complex [36–38]. Conversely, in the searching scenario with a narrow range, CEFs based on spatial domain information will be overly smoothed, while frequency-based CEFs can offer a higher axial resolution.

The features of CEFs are contingent upon the sample type and focusing scenarios. A CEF is tailored to specific characteristics that apply to a particular scenario, rendering them non-adjustable. A universal and adaptable CEF is necessary to overcome this limitation, capable of conforming to any sample type or focusing scheme.

2.2. Flexible and universal CEF

The process of evaluating image clarity involves segregating samples into two categories – binary and continuous samples. Binary samples comprise amplitude images, where spatial amplitude is the primary factor for determining clarity. Continuous samples, on the other hand, include transparent or partially transparent objects, such as stained and unstained cells, which have non-uniform distributions of refractive index or thickness, leading to significant phase changes in the transmitted light. Consequently, evaluating clarity in these cases necessitates an analysis of the frequency domain. Since image clarity depends on spatial and frequency domain characteristics, and different samples require different evaluation criteria, we recommend incorporating both domains in sample images.

The Fourier Transform is a linear transformation method that converts a signal from the spatial domain to the frequency domain, providing the overall frequency spectrum of the signal. It can be viewed as rotating the signal $\pi/2$ from the spatial domain to the frequency domain. In contrast, the FrFT allows for the rotation of a signal between the spatial and frequency domains in a fractional domain, with the rotation angle determined by the fractional order of the FrFT [39–41]. By analyzing the FrFT spectrum of an image, we can obtain information about both its spatial and frequency domains simultaneously.

R. Li et al. have proposed a qualitative edge detection method utilizing FrFT [42]. To obtain quantitative results, we develop an image clarity evaluation algorithm based on the amplitude difference of the FrFT (ADFrFT). This approach employs the fractional order p as a flexible parameter, introducing a new dimension of control to the algorithm. The process involves performing p -order FrFT on each plane's complex amplitude separately, generating fractional spectra. Afterward, a $(p - \Delta p)$ -order inverse FrFT (iFrFT) is applied to the fractional spectra, creating another set of complex amplitudes. We can obtain quantitative information about the image's edges by computing the difference between these two sets of complex amplitudes. Assuming $I(x, y; z)$ has $L_x \times L_y$ pixels, ADFrFT can be expressed as:

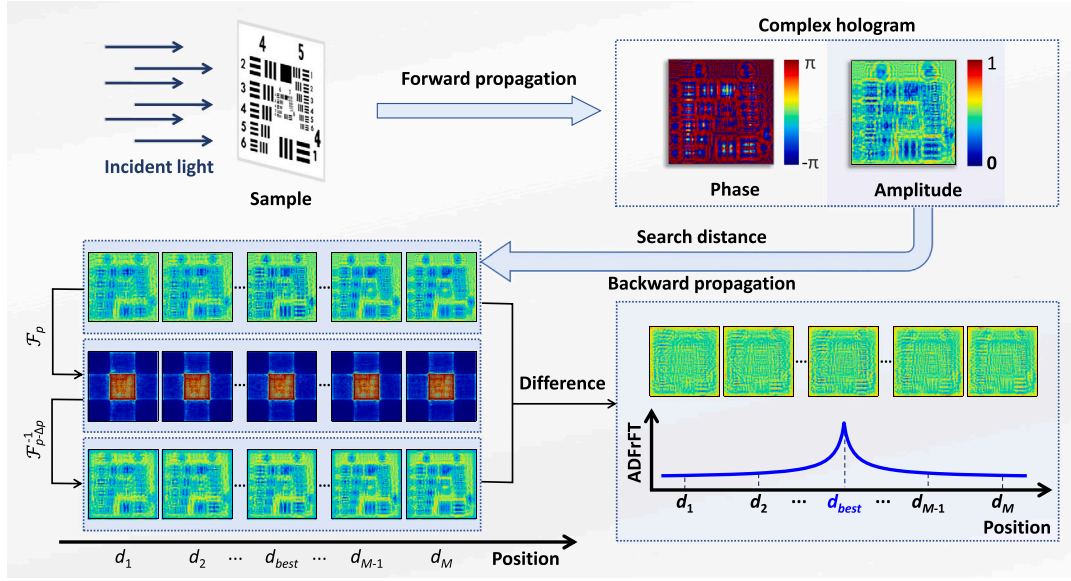


Fig. 2. Principle of autofocus based on fractional Fourier amplitude difference information.

$$\text{ADFrFT}(z_n, p, \Delta p) = \frac{1}{L_x L_y} \sum_{x=1}^{L_x} \sum_{y=1}^{L_y} \left| \mathcal{F}_p^{-1} \left\{ \mathcal{F}_p \left[I_n(x, y; z_n) \right] \right\} \right| + (\Delta p - 1) I_n(x, y; z_n) \quad (1)$$

where $\mathcal{F}_p$ denotes p -order FrFT, and $\mathcal{F}_{p-\Delta p}^{-1}$ denotes $(p - \Delta p)$ -order iFrFT. The first item of Eq. (1) represents the fractional Fourier amplitude difference of $I_n(x, y; z_n)$, and the second item represents the amplitude compensation. Eq. (1) can be mathematically simplified as:

$$\text{ADFrFT}(z_n, \Delta p) = \frac{1}{L_x L_y} \sum_{x=1}^{L_x} \sum_{y=1}^{L_y} \left| \mathcal{F}_{\Delta p} \left[I_n(x, y; z_n) \right] \right| + (\Delta p - 1) I_n(x, y; z_n) \quad (2)$$

Eq. (2) reveals that the value of ADFrFT remains unaffected by p , and instead depends solely on the difference in orders between the two FrFT operations Δp . The range of Δp lies between 0 and 1. The principle is illustrated in Fig. 2. A small Δp implies that the spatial domain of the sample has a greater impact on ADFrFT, whereas a large Δp gives the frequency domain a greater influence. When Δp assumes an intermediate value, ADFrFT combines spatial and frequency domain information by utilizing the fractional domain between them. Compared to traditional CEFs, ADFrFT offers an additional degree of freedom, which enables the autofocus function to adjust to various sample types and focusing scenarios. The ensuing sections will delve deeper into the flexibility that this degree of freedom provides and the algorithm's universality.

3. Simulations and experiments

3.1. Advantages of ADFrFT

To assess the effectiveness of ADFrFT in evaluating image clarity across various sample types, we utilized both a binary sample in the form of the positive USAF 1951 target, and a continuous sample featuring Guinea pig hair shaft infected with trichophyton mentagrophytes (HSTM), as depicted in Fig. 3. The difference between the two types of samples is reflected in the modulation effect on the complex holographic light field and the degree of absorption. The binary positive USAF 1951 target is a typical amplitude-type sample, while transparent or partially transparent samples such as HSTM can be utilized as phase-type samples. To demonstrate the noise robustness of the new method, we introduced Gaussian noise with a variance of 0.05 to both samples,

resulting in the images displayed in Fig. 3(c, d). We apply Fig. 3(a–d) as the sample to carry out numerical simulations. The center wavelength of the laser is 532 nm, with the sample located 5 mm from the detector. We extracted the intensity distribution of the complex holographic light field at the detector plane and utilized a traversal method with a precision of 0.01 mm to pinpoint the diffraction distance. The resulting normalized clarity varies with both propagation distance and Δp can be observed in Fig. 3(a1–d1).

The position where the normalized clarity reaches its maximum value indicates the focus has been reached, and the corresponding horizontal coordinate corresponds to the distance from the detector to the sample. The ADFrFT method can accurately determine the propagation distance of various sample types by utilizing different values of Δp . When we need a smooth autofocus curve, or the spatial domain of the sample is more significant, we set Δp to a smaller value. Conversely, when we need an autofocus curve with higher axial resolution or the frequency domain of the sample is more complex, we set Δp to a larger value. According to Fig. 2(a1–d1), in order to show representative results, we set $\Delta p = 0.2$ and 0.8 , respectively. The autofocus curve with $\Delta p = 0.2$ has the general characteristics of spatial-based CEFs, while $\Delta p = 0.8$ has the general characteristics of frequency-based CEFs.

As shown in Fig. 3(a2–b2), it is evident that when Δp is smaller, ADFrFT prioritizes spatial domain information, resulting in smoother autofocus curves with fewer occurrences of local extrema. On the other hand, as Δp increases, ADFrFT places greater importance on frequency domain information. A Δp value of 0.8 significantly improves the kurtosis of the autofocus curve, leading to enhanced axial resolution. Furthermore, as shown in Fig. 3(c2–d2), the introduction of Gaussian noise was discovered to enhance the axial resolution. This is due to the fact that the addition of noise mimics the effect of scatter illumination on the sample. The unique local scatter is globally significant in relation to the whole projected scatter pattern. It can also prevent the negative impact of aliasing artifacts on the focusing search [26]. However, not all CEFs have ideal results for speckle noise, such as ToG and GoG. ADFrFT exhibits strong noise robustness as it can accurately locate sample positions even after introducing noise.

Autofocus curves featuring different Δp values exhibit distinct features that demonstrate the efficacy and flexibility of ADFrFT in providing additional degrees of freedom. Choosing the optimal Δp for the sample type and focus scenario can result in more suitable autofocus curves, a capability that other CEFs cannot provide.

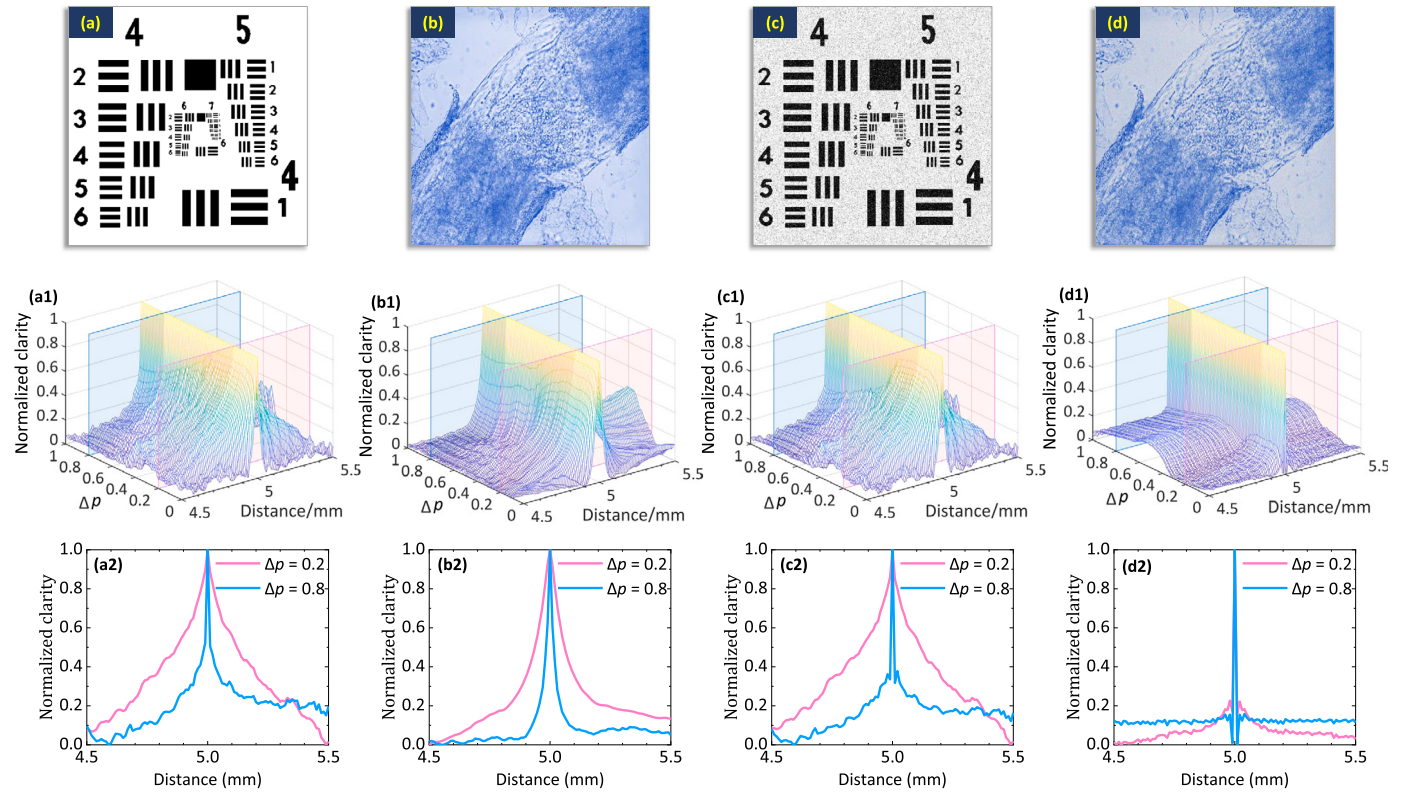


Fig. 3. Different types of samples and corresponding normalized clarity calculated by ADFrFT. (a) the positive USAF 1951 target. (b) Guinea pig hair shaft infected with the trichophyton mentagrophytes [43]. (c) and (d) have been added Gaussian noise with a variance of 0.05 corresponding to (a) and (b) respectively. (a1–d1) is the normalized clarity varies with distance and Δp . (a2–d2) corresponds to the autofocus curves at $\Delta p = 0.2$ and $\Delta p = 0.8$.

Table 1.A

FWHM calculation results of different CEFs for each sample. (Unit: mm).

	GRA	LAP	SG	SPEC	ToG	GoG	Brenner	VAR	AMP	Tenengrad	ADFrFT $\Delta p = 0.2$	ADFrFT $\Delta p = 0.8$
USAF 1951	/	0.033	0.048	/	0.051	0.066	0.046	0.241	0.241	0.058	0.251	0.025
USAF 1951-N-O	0.012	0.014	0.014	0.020	0.051	0.053	0.014	0.126	0.126	0.033	0.142	0.010
USAF 1951-N-I	0.112	0.045	0.068	/	0.043	0.047	0.069	0.283	0.283	0.089	0.308	0.105
HSTM	0.053	0.037	0.051	0.010	0.072	0.074	0.047	0.118	0.118	0.058	0.124	0.033
HSTM-N-O	0.010	0.012	0.010	0.010	/	/	0.010	0.012	0.012	0.021	0.012	0.010
HSTM-N-I	0.052	0.038	0.051	/	0.075	0.060	0.048	0.123	0.123	0.059	0.129	0.052

Table 1.B

UR calculation results of different CEFs for each sample. (Unit: mm).

	GRA	LAP	SG	SPEC	ToG	GoG	Brenner	VAR	AMP	Tenengrad	ADFrFT $\Delta p = 0.2$	ADFrFT $\Delta p = 0.8$
USAF 1951	/	0.13	0.16	/	0.26	0.36	0.21	0.48	0.48	0.16	0.50	0.14
USAF 1951-N-O	0.03	0.07	0.10	0.02	0.12	0.17	0.10	0.37	0.37	0.16	0.49	0.02
USAF 1951-N-I	0.08	0.14	0.14	/	0.16	0.14	0.14	0.31	0.31	0.15	0.31	0.16
HSTM	0.26	0.16	0.27	0.02	0.35	0.35	0.27	0.70	0.70	0.27	0.75	0.24
HSTM-N-O	0.06	0.06	0.06	0.02	/	/	0.06	0.02	0.02	0.07	0.02	0.02
HSTM-N-I	0.16	0.11	0.17	/	0.13	0.07	0.17	0.32	0.32	0.26	0.33	0.16

3.2. Comparison of the CEFs

We conducted a comparative study to showcase the advantages of ADFrFT over various commonly used CEFs with Δp values of 0.2 and 0.8. In practical scenarios, noise can be present not only in the object plane but also in the image plane. Therefore, the performance of various CEFs is evaluated by adding Gaussian noise with a variance of 0.05 to the object plane and the image plane. The results are presented in Fig. 4.

To quantitatively evaluate the performance of various CEFs, the assessment includes measuring the full width at half maximum (FWHM), the unimodal range (UR), the number of false maxima (NFM), and the clarity difference between the maximum and the secondary maximum

(CDBM). FWHM indicates the sharpness of the CEF, with a smaller FWHM indicating higher axial resolution, and a larger FWHM providing better search capabilities for a larger depth of field. UR is defined as the distance between two adjacent local minima neighboring the central peak. A wider UR can mitigate potential errors caused by local extrema. NFM represents the number of maxima, excluding the central peak, and a lower NFM value indicates better stability. In experiments, sub-peaks interfere with the central peak, and a larger CDBM means less interference from the sub-peak signal.

The quantitative results from Fig. 4 are summarized in Table 1.A–1.D. ADFrFT consistently obtains accurate results for all types of samples. After introducing noise to both the object plane and image

Table 1.C

NFM calculation results of different CEFs for each sample.

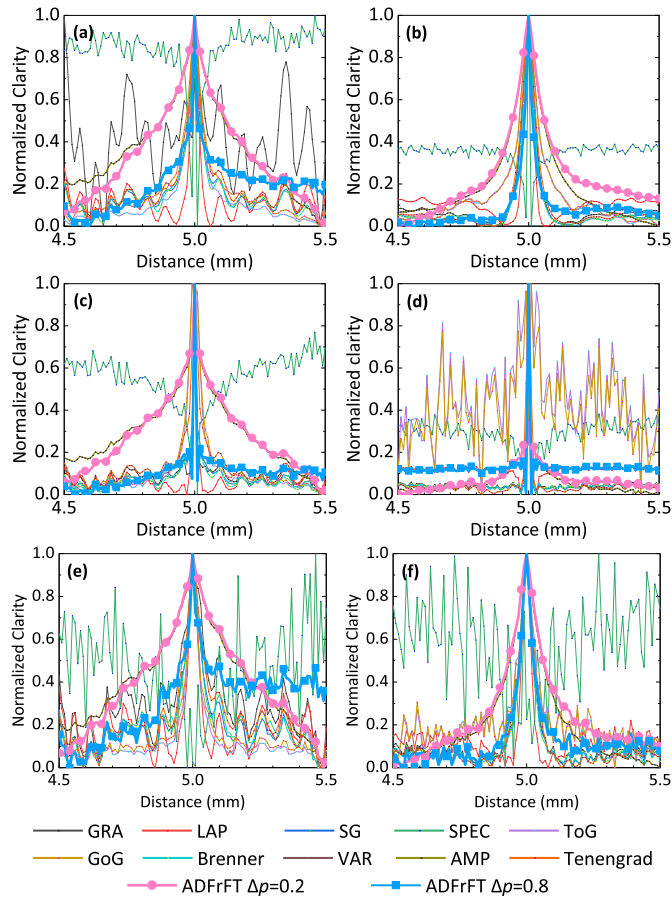
	GRA	LAP	SG	SPEC	ToG	GoG	Brenner	VAR	AMP	Tenengrad	ADFrFT $\Delta p = 0.2$	ADFrFT $\Delta p = 0.8$
USAF 1951	/	11	11	/	11	11	10	7	7	11	7	22
USAF 1951-N-O	19	16	15	29	19	20	16	7	7	11	7	31
USAF 1951-N-I	18	18	19	/	19	23	21	15	15	11	10	28
HSTM	12	11	11	29	9	15	10	3	3	10	1	17
HSTM-N-O	31	30	31	30	/	/	32	30	30	20	29	35
HSTM-N-I	27	33	30	/	25	30	28	19	19	14	18	28

Table 1.D

CDBM calculation results of different CEFs for each sample.

	GRA	LAP	SG	SPEC	ToG	GoG	Brenner	VAR	AMP	Tenengrad	ADFrFT $\Delta p = 0.2$	ADFrFT $\Delta p = 0.8$
USAF 1951	/	0.723	0.803	/	0.872	0.843	0.773	0.552	0.552	0.735	0.821	0.676
USAF 1951-N-O	0.841	0.879	0.830	0.240	0.040	0.862	0.845	0.751	0.751	0.801	0.802	0.784
USAF 1951-N-I	0.483	0.659	0.691	/	0.863	0.842	0.633	0.533	0.534	0.655	0.612	0.519
HSTM	0.912	0.860	0.946	0.613	0.873	0.880	0.938	0.935	0.935	0.934	0.861	0.909
HSTM-N-O	0.935	0.857	0.942	0.619	/	/	0.941	0.772	0.772	0.947	0.786	0.909
HSTM-N-I	0.863	0.769	0.872	/	0.649	0.654	0.881	0.827	0.828	0.891	0.772	0.769

N indicates samples have been added noise; O indicates the object plane; I indicates the image plane; / indicates CEF failure.

**Fig. 4.** Autofocus curves comparisons for various CEFs, including commonly used CEFs and ADFrFT. (a) corresponds to the positive USAF 1951 target, (b) corresponds to the HSTM, (c) and (d) correspond to (a) and (b) respectively, with Gaussian noise added to the object plane, while (d) and (e) correspond to (a) and (b) respectively, with Gaussian noise added to the image plane.

plane, the results remain highly accurate. When Δp is set to a smaller value, it corresponds to wider FWHM and UR values, with the fewest NFM, resulting in smoother curves. Conversely, when Δp is set to a larger value, it corresponds to narrower FWHM values, indicating more substantial axial resolution. When the search range is unknown and the

initial range does not include the focus, CEFs with wider FWHM can provide guidance for the search, which is applicable to the initial guess of the defocusing distance to optimize the search range in the right direction. We can use an autofocus strategy that starts with sparse sampling using FWHM-wide CEFs to find the range of the initial position and then uses FWHM-narrow CEFs for precise localization. The FWHM of ADFrFT can be switched between wide and narrow, therefore only ADFrFT is required to implement this strategy. Although the CDBM of ADFrFT is not the largest, it also manages to consistently maintain a high value, performing well and stable compared to other CEFs.

Furthermore, it is important to note that the selected values of $\Delta p = 0.2$ and $\Delta p = 0.8$ were utilized for illustrative purposes to showcase the influence of the degree of freedom on ADFrFT. In practical scenarios, the optimal choice of Δp can be tailored to meet specific requirements and utilize prior knowledge to better suit the characteristics of the given sample, ultimately achieving optimal performance in image clarity evaluation and autofocus functionality. This highlights the noteworthy flexibility and universality of ADFrFT compared to other CEFs in a variety of sample types and autofocus scenarios.

3.3. Autofocus experimental results

We conducted experiments using the system setup illustrated in Fig. 1. The sample received illumination from a fiber laser with a center wavelength of 532 nm. A CMOS detector (DMK27AUJ003, pixel pitch is 1.67 μm) was used to capture the intensity dataset of multi-distance diffraction images.

The positive USAF 1951 target experiment was conducted, and we recorded diffraction images at four different diffraction planes. Based on simulation results, we selected $\Delta p = 0.2$ and $\Delta p = 0.8$ for ADFrFT. Fig. 5 depicts the diffraction images and their corresponding autofocus curves. To determine the accuracy of the autofocus results, we manually refocused on different planes and selected the best one, which was the most accurate but time-consuming method. The manually accurate results are marked with a black dashed line. GRA and SPEC failed to identify the correct diffraction distance, while SG, Brenner, Tenengrad, and other curves exhibited significant fluctuations and poor sensitivity. Among all the CEFs, ADFrFT with $\Delta p = 0.2$ demonstrated the widest FWHM and UR, the smoothest performance, and the least NFM. ADFrFT with $\Delta p = 0.8$ has a better resolution. As a result, it is more suitable for algorithm-based intelligent searching methods.

We have replaced the previous sample with a histopathological section of thyroid adenocarcinoma (HSTA). The image contains various types of detailed information, and in order to produce optimal imaging

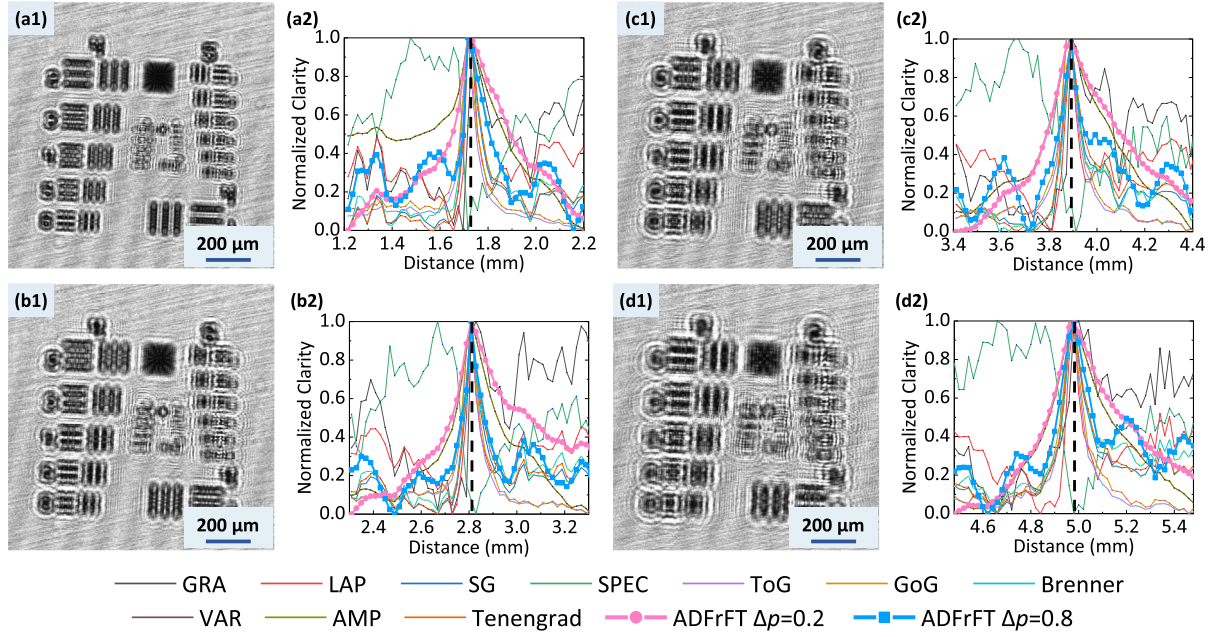


Fig. 5. The autofocus results of the positive USAF 1951 target. (a1–d1) show the diffraction images of the light field at four different propagation distances, and (a2–d2) correspond to their respective autofocus curves.

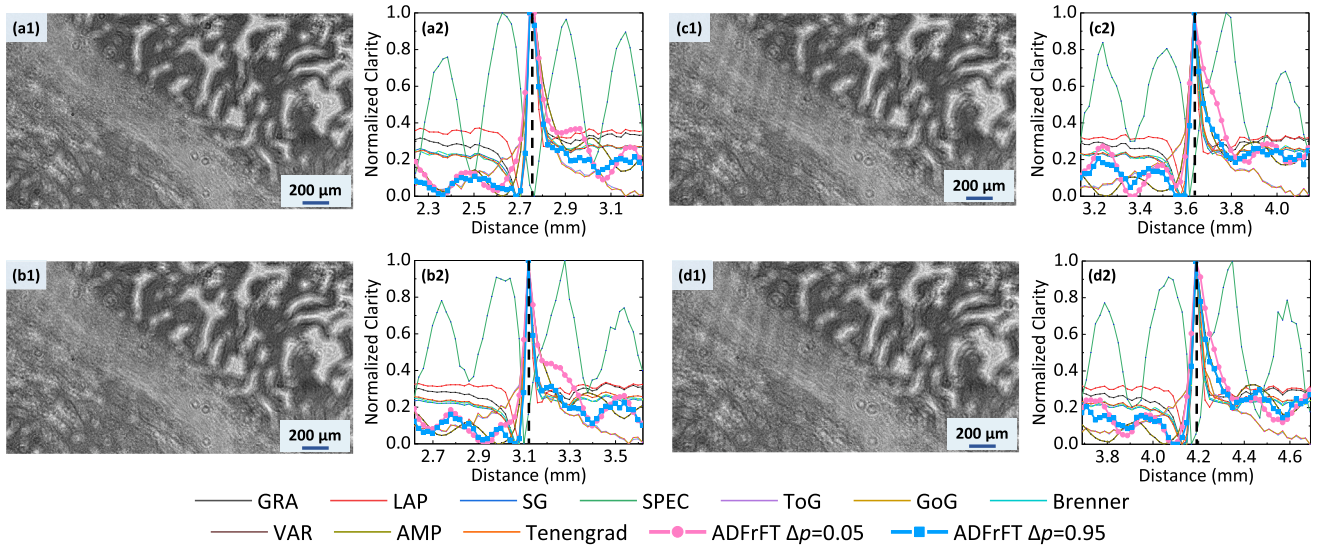


Fig. 6. The autofocus results of the HSTA. (a1–d1) show the diffraction images of the light field at four different propagation distances, and (a2–d2) correspond to their respective autofocus curves.

results, it is crucial to prioritize the accuracy of the diffraction distance search. This is reflected in the higher axial resolution and minor fluctuations in the autofocus curve, which can be achieved by emphasizing frequency domain information and setting $\Delta p = 0.95$. To reflect the flexibility of the ADFrFT accordingly, $\Delta p = 0.05$ was chosen as the comparison. The corresponding diffraction images and autofocus curves are presented in Fig. 6. Unfortunately, SPEC did not meet our expectations in this particular case. However, ADFrFT proved to be highly effective and produced sharp peaks similar to LAP, SG, and other CEFs, but with fewer NFM. While fluctuations in GRA, LAP, SG, and other CEFs tend to increase as distance increases, ADFrFT is less susceptible to these changes. It delivers reliable outcomes and enables adjusting of the FWHM through an additional degree of freedom. Compared to other CEFs, ADFrFT exhibits more remarkable universality and adaptability, making it more advantageous for practical applications.

3.4. MDCDI experimental results

Within this section, we will employ ADFrFT to coaxial MDCDI to confirm the precision of the outcomes and enhance the performance of the multi-distance phase retrieval algorithm for object reconstruction. The autofocus findings will enable us to refine the accuracy of our physical distance assessments. We utilized multi-distance diffraction images and precisely positioned diffraction distances as inputs to engage the multi-distance phase retrieval algorithm for object reconstruction. Ultimately, we compared the reconstructed outcomes obtained through the physical ranging method with those obtained via our suggested ADFrFT. The results of physical ranging and ADFrFT positioning are shown in Table 2.

Fig. 7(a1) illustrates the reconstruction of the positive USAF 1951 target through the physical ranging method, while Fig. 7(b1) uses ADFrFT positioned distance. Upon closer inspection of the magnified

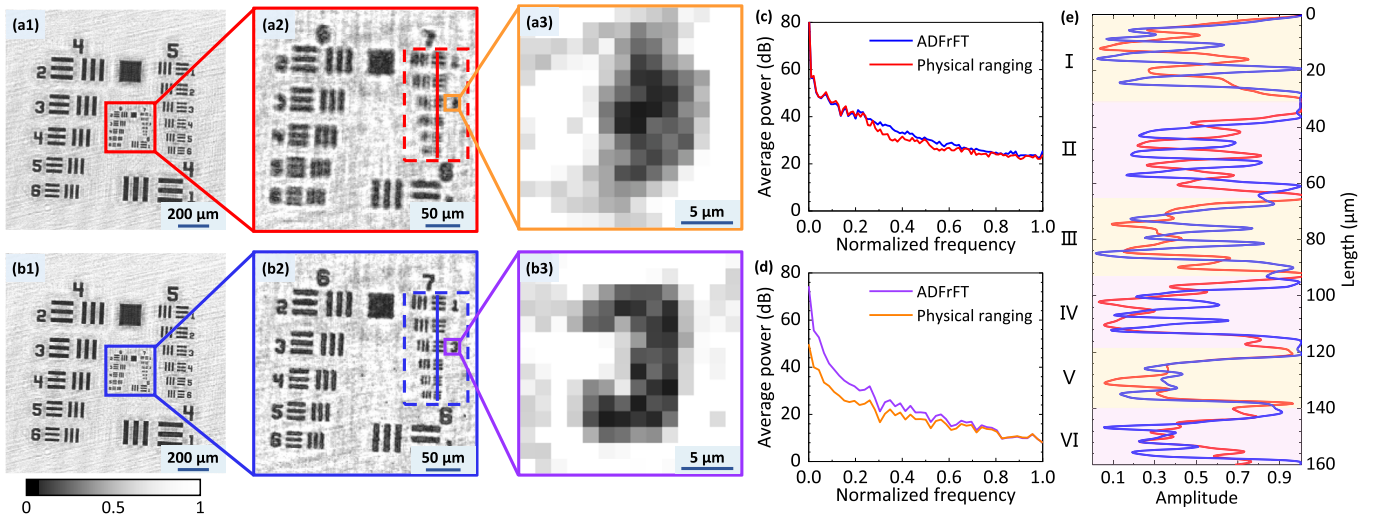


Fig. 7. The MDCDI results of the positive USAF 1951 target. (a1) corresponds to the reconstructed results using the physical ranging method, (b1) corresponds to the reconstructed results using the distance positioned by ADFrFT, (a2) and (b2) correspond to the magnified central region, (a3) and (b3) correspond to the imaging results of number 3. (c) compares the one-dimensional power spectrum curve of (a2) and (b2), and (d) compares the power spectrum of (a3) and (b3). The line-scan profiles are shown in (e), where the red line represents the physical ranging method and the blue line represents the ADFrFT positioned method. The line pairs in the 1st through 6th ranges are designated by numbers from I to VI.

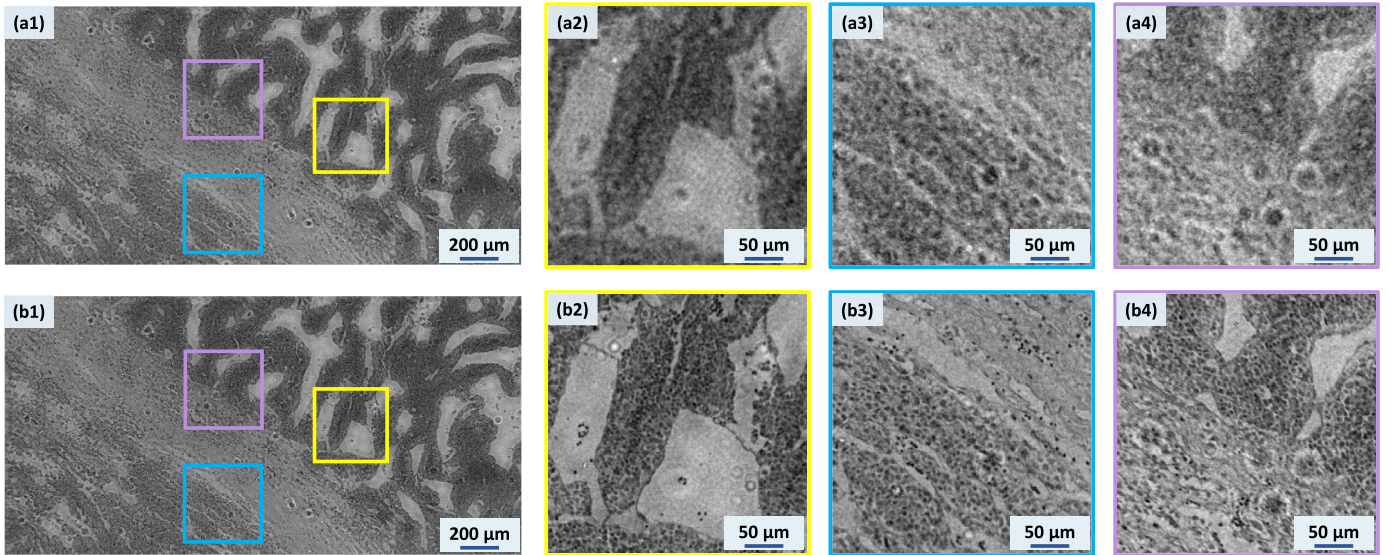


Fig. 8. The MDCDI results of the HSTA. (a1) corresponds to the reconstructed results using the physical ranging method, and (b1) corresponds to the reconstructed results using the distance positioned by ADFrFT. (a2–a4) are three different magnified regions of (a1) with different tissue structures, and (b2–b4) correspond to the same magnified regions of (b1).

Table 2

Comparison of distance measurement results. (Unit: mm).

		z_1	z_2	z_3	z_4
USAF 1951	Physical ranging	1.72	2.81	3.90	4.96
	ADFrFT	1.716	2.809	3.912	4.980
HSTA	Physical ranging	2.78	3.09	3.50	4.21
	ADFrFT	2.744	3.118	3.637	4.188

central region in Fig. 7(a2), we can observe that it lacks diffraction distance accuracy, which results in decreased imaging resolution. In contrast, Fig. 7(b2) presents an updated reconstruction that previously appeared indistinguishable, displaying clear and distinguishable numbers and lines. In the selected area of Fig. 7(a2) and (b2), imaging results of number 3 are shown in Fig. 7(a3) and (b3), respectively. Fig. 7(a3) is difficult to recognize, while Fig. 7(b3) boasts a high con-

trast ratio. Fig. 7(c) displays the power spectrum curve of Fig. 7(a2) and (b2). The average power of ADFrFT exceeds that of the physical ranging method. Fig. 7(d) showcases a comparison of Fig. 7(a3) and (b3), revealing that the power spectrum of ADFrFT is superior to the physical ranging method at any spatial frequency, providing proof that Fig. 7(b3) boasts better quality. Fig. 7(e) presents line-scan profiles from the selected area, with the physical ranging method represented by the red line and the ADFrFT positioned method represented by the blue line. Ideally, the profile should have three peaks with amplitudes equal to 0 within each line pair range. The physical distance measurement method can accurately resolve up to the 3rd line pair in the 7th group with an accuracy of 161 lp/mm, while the ADFrFT method can resolve the 6th line pair in the 7th group with an accuracy of 228 lp/mm. Furthermore, the ADFrFT method can reveal the peaks from the 4th to 6th pair, which are indistinguishable using physical ranging methods. Therefore,

we have demonstrated a qualitative and quantitative improvement in coaxial MDCDI by using ADFrFT.

Fig. 8 presents the experimental results for the HSTA. Fig. 8(a1) displays the reconstruction result based on the physical ranging method, while Fig. 8(b1) displays the reconstruction result based on the ADFrFT. Fig. 8(a2, a3, a4) and (b2, b3, b4) correspond to local enlargements of three different tissue structures. It is clear that the reconstructed image has low resolution and cannot display numerous details when the accuracy of diffraction distance is inadequate. On the other hand, more precise sample diffraction distances obtained through ADFrFT can significantly enhance the resolution of the reconstructed images. This approach eliminates the need for high-precision distance measurement devices, simplifies the experimental setup, and enables high-resolution images to be obtained without the need for post-processing algorithms, thereby improving imaging efficiency.

In this paper, we have utilized ADFrFT on MDCDI. However, it is worth noting that this algorithm is not limited to this particular context. Similar to conventional CEFs, ADFrFT can be employed in a range of computational imaging applications, such as digital holography to ascertain object distance and defocus, as well as autofocus in optical microscopy systems.

4. Conclusion

We propose an autofocus algorithm that utilizes the amplitude difference between adjacent fractional domains to construct a CEF. This CEF is used to calculate the distance between the detector plane and the sample plane or assess the defocus condition of the light field within a specific range. ADFrFT demonstrates superior flexibility and universality compared to other CEFs. Proper selection of Δp values allows for accurate distance searches to samples of any type with robustness against noise, validated by MDCDI experimental results. The proposed method achieves higher resolution than physical measurement.

Funding

This work was supported by the National Natural Science Foundation of China (Nos. 61975044, 12074094, 12374271, 62375065); Fundamental Research Funds for the Central Universities (No. HIT-DIJJ.2023175); Interdisciplinary Research Foundation of HIT (No. IR2021237).

CRedit authorship contribution statement

Ziyang Li: Conceptualization, Investigation, Methodology, Writing – original draft. **Xuyang Zhou:** Writing – review & editing. **Yiran Wang:** Investigation. **Guancheng Huang:** Investigation. **Shutian Liu:** Writing – review & editing. **Bin Gao:** Writing – review & editing. **Zhengjun Liu:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

References

- [1] Li M, Bian L, Zhang J. Multi-slice coded coherent diffraction imaging. *Opt Lasers Eng* 2022;151. <https://doi.org/10.1016/j.optlaseng.2021.106929>.
- [2] Liu Y, Liu Q, Li Y, Zhang J, He Z. High-resolution multi-planar coherent diffraction imaging with multimode fiber source. *Opt Lasers Eng* 2021;140. <https://doi.org/10.1016/j.optlaseng.2021.106530>.
- [3] Kuzmenko AV, Butok OM. Phase retrieval for arbitrary complex-valued objects using structured illumination. *Opt Express* 2023;31(15):24505–15. <https://doi.org/10.1364/OE.493331>.
- [4] Li X, Liu K, Liu Z, Guo J. High-precision phase retrieval method for speckle suppression based on optimized modulation masks. *Opt Express* 2023;31(12):18824–39. <https://doi.org/10.1364/OE.489492>.
- [5] Miao J, Ishikawa T, Robinson IK, Murnane MM. Beyond crystallography: diffractive imaging using coherent x-ray light sources. *Science* 2015;348(6234):530–5. <https://doi.org/10.1126/science.aaa1394>.
- [6] Zheng G, Horstmeyer R, Yang C. Wide-field, high-resolution Fourier ptychographic microscopy (vol 7, pg 739, 2013). *Nat Photonics* 2015;9(9):621. <https://doi.org/10.1038/NPHOTON.2015.148>.
- [7] Jiang S, Song P, Wang T, Yang L, Wang R, Guo C, et al. Spatial- and Fourier-domain ptychography for high-throughput bio-imaging. *Nat Protoc* 2023;18(7):2051–83. <https://doi.org/10.1038/s41596-023-00829-4>.
- [8] Yin D, Gu Z, Zhang Y, Gu F, Nie S, Feng S, et al. Speckle noise reduction in coherent imaging based on deep learning without clean data. *Opt Lasers Eng* 2020;133. <https://doi.org/10.1016/j.optlaseng.2020.106151>.
- [9] Pfeiffer F. X-ray ptychography. *Nat Photonics* 2018;12(1):9–17. <https://doi.org/10.1038/s41566-017-0072-5>.
- [10] Luo W, Zhang Y, Feizi A, Gorocs Z, Ozcan A. Pixel super-resolution using wavelength scanning. *Light: Sci Appl* APR 2016;5. <https://doi.org/10.1038/lsa.2016.60>.
- [11] Uo C, Iang S, Ong P, Wang T, Shao X, Zhang Z, et al. Quantitative multi-height phase retrieval via a coded image sensor. *Biomed Opt Express* 2021;12(11):7173–84. <https://doi.org/10.1364/BOE.443528>.
- [12] Jiang S, Guo C, Hu P, Hu D, Song P, Wang T, et al. High-throughput lensless whole slide imaging via continuous height-varying modulation of a tilted sensor. *Opt Lett* 2021;46(20):5212–5. <https://doi.org/10.1364/OL.437832>.
- [13] Wen X, Zhou X, Li Y, Ji Y, Zhou K, Liu S, et al. High-performance lensless diffraction imaging from diverse holograms by three-dimensional scanning. *Opt Lett* 2022;47(14):3423–6. <https://doi.org/10.1364/OL.464864>.
- [14] Rivenson Y, Zhang Y, Gnyadin H, Teng D, Ozcan A. Phase recovery and holographic image reconstruction using deep learning in neural networks. *Light: Sci Appl* 2018;7. <https://doi.org/10.1038/lsa.2017.141>.
- [15] Guo C, Zhao Y, Tan J, Liu S, Liu Z. Multi-distance phase retrieval with a weighted shrink-wrap constraint. *Opt Lasers Eng* 2019;113:1–5. <https://doi.org/10.1016/j.optlaseng.2018.09.017>.
- [16] Bao Y, Sun L, Ma Y, Gu D, Wang K. Improving fast auto-focus with event polarity. *Opt Express* 2023;31(15):24025–44. <https://doi.org/10.1364/OE.489717>.
- [17] Zhang Q, Wang Y, Li Q, Tao X, Zhou X, Zhang Y, et al. An autofocus algorithm considering wavelength changes for large scale microscopic hyperspectral pathological imaging system. *J Biophotonics* 2022;15(5). <https://doi.org/10.1002/jbio.202100366>.
- [18] Bian Z, Guo C, Jiang S, Zhu J, Wang R, Song P, et al. Autofocusing technologies for whole slide imaging and automated microscopy. *J Biophotonics* 2020;13(12). <https://doi.org/10.1002/jbio.202000227>.
- [19] Wu X, Zhou H, Yu H, Hu R, Zhang G, Hu J, et al. A method for medical microscopic images' sharpness evaluation based on NSST and variance by combining time and frequency domains. *Sensors* 2022;22(19). <https://doi.org/10.3390/s22197607>.
- [20] Silvestri L, Mullenbroich MC, Costantini I, Di Giovanna AP, Mazzamuto G, Franceschini A, et al. Universal autofocus for quantitative volumetric microscopy of whole mouse brains. *Nat Methods* 2021;18(8):953–8. <https://doi.org/10.1038/s41592-021-01208-1>.
- [21] Xia X, Yao Y, Liang J, Fang S, Yang Z, Cui D. Evaluation of focus measures for the autofocus of line scan cameras. *Optik* 2016;127(19):7762–75. <https://doi.org/10.1016/j.jpleo.2016.05.114>.
- [22] Xia X, Yin L, Yao Y, Wu W, Zhang Y. Combining two focus measures to improve performance. *Meas Sci Technol* OCT 2017;28. <https://doi.org/10.1088/1361-6501/aa7fe2>.
- [23] Trusiak M, Picazo-Bueno J-A, Zdankowski P, Mico V. Darkfocus: numerical auto-focusing in digital in-line holographic microscopy using variance of computational dark-field gradient. *Opt Lasers Eng* 2020;134. <https://doi.org/10.1016/j.optlaseng.2020.106195>.
- [24] Zhou X, Wen X, Ji Y, Li Y, Liu S, Liu Z. Image reconstruction using autofocus in single-lens system. *Appl Sci* 2022;12(3). <https://doi.org/10.3390/app12031378>.
- [25] He C, Li X, Hu Y, Ye Z, Kang H. Microscope images automatic focus algorithm based on eight-neighborhood operator and least square planar fitting. *Optik* 2020;206. <https://doi.org/10.1016/j.jpleo.2020.164232>.
- [26] Guo C, Zhao Y, Tan J, Liu S, Liu Z. Adaptive lens-free computational coherent imaging using autofocusing quantification with speckle illumination. *Opt Express* 2018;26(11):14407–20. <https://doi.org/10.1364/OE.26.014407>.
- [27] Yang C, Chen M, Zhou F, Li W, Peng Z. Accurate and rapid auto-focus methods based on image quality assessment for telescope observation. *Appl Sci* 2020;10(2). <https://doi.org/10.3390/app10020658>.
- [28] Zhang X, Wu H, Ma Y. A new auto-focus measure based on medium frequency discrete cosine transform filtering and discrete cosine transform. *Appl Comput Harmon Anal* 2016;40(2):430–7. <https://doi.org/10.1016/j.acha.2015.08.008>.

- [29] Memmolo P, Distanto C, Paturzo M, Finizio A, Ferraro P, Javidi B. Automatic focusing in digital holography and its application to stretched holograms. *Opt Lett* 2011;36(10):1945–7. <https://doi.org/10.1364/OL.36.001945>.
- [30] Memmolo P, Paturzo M, Javidi B, Netti PA, Ferraro P. Refocusing criterion via sparsity measurements in digital holography. *Opt Lett* 2014;39(16):4719–22. <https://doi.org/10.1364/OL.39.004719>.
- [31] Zhang Y, Wang H, Wu Y, Tamamitsu M, Ozcan A. Edge sparsity criterion for robust holographic autofocusing. *Opt Lett* 2017;42(19):3824–7. <https://doi.org/10.1364/OL.42.003824>.
- [32] Zhou X, Xiong P, Chi D, Wen X, Ji Y, Li Y, et al. Fast autofocusing based on pixel difference with the Tanimoto coefficient between images. *Opt Lett* 2022;47(15):3752–5. <https://doi.org/10.1364/OL.463482>.
- [33] Wen X, Li Y, Zhou X, Ji Y, Zhou K, Liu S, et al. Phase retrieval with dynamic linear combination in multiple intensity measurements. *Opt Lasers Eng* 2022;159. <https://doi.org/10.1016/j.optlaseng.2022.107200>.
- [34] Langehanenberg P, Kemper B, Dirksen D, von Bally G. Autofocusing in digital holographic phase contrast microscopy on pure phase objects for live cell imaging. *Appl Opt* 2008;47(19):D176–82. <https://doi.org/10.1364/AO.47.00D176>.
- [35] Sun Y, Duthaler S, Nelson B. Autofocusing in computer microscopy: selecting the optimal focus algorithm. *Microsc Res Tech* 2004;65(3):139–49. <https://doi.org/10.1002/jemt.20118>.
- [36] Rathod S, Ghosh A, Kulkarni R. Fast and accurate autofocusing algorithm in digital holography based on particle swarm optimization. *Optik* 2021;247. <https://doi.org/10.1016/j.ijleo.2021.167946>.
- [37] Wang G, Huang J, Lyu J, Chen Z, Lu R, Cao Z, et al. Fast autofocusing algorithm based on variable step search and curve fitting applied in the near- and far-field detection system with a common optical path. *J Lightwave Technol* 2023;41(6):1742–8. <https://doi.org/10.1109/JLT.2022.3225545>.
- [38] Zhou X, Wen X, Ji Y, Geng Y, Liu S, Liu Z. Fast automatic multiple positioning for lensless coherent diffraction imaging. *Opt Lasers Eng* 2022;155. <https://doi.org/10.1016/j.optlaseng.2022.107055>.
- [39] Bahri M, Abdul Karim SA. Fractional Fourier transform: main properties and inequalities. *Mathematics* MAR 2023;11(5). <https://doi.org/10.3390/math11051234>.
- [40] Kumar S, Saxena R, Singh K. Fractional Fourier transform and fractional-order calculus-based image edge detection. *Circuits Syst Signal Process* 2017;36(4):1493–513. <https://doi.org/10.1007/s00034-016-0364-x>.
- [41] Ozaktas H, Ankan O, Kutay M, Bozdagi G. Digital computation of the fractional Fourier transform. *IEEE Trans Signal Process* 1996;44(9):2141–50. <https://doi.org/10.1109/78.536672>.
- [42] Li R, Hong J, Zhou X, Li Q, Zhang X. Fractional Fourier single-pixel imaging. *Opt Express* 2021;29(17):27309–21. <https://doi.org/10.1364/OE.434103>.
- [43] Ajello L. Public health image library (PHIL). <https://phil.cdc.gov/Details.aspx?pid=22121>, 1964.